Version 1 assumptions: analysis and sensitivity

Which assumptions drive the result, which are justified, and what to measure to calibrate the model

Lambach Experimental Aircraft · branch **optimization** · 13 Nov 2026

Main findings

1. **The drag model is the biggest uncertainty.** Adding plausible motor/nacelle drag doubles the predicted drag (+106 %), and fully turbulent flow on the lifting surfaces adds 25 %. These terms must be measured before anything else is tuned.
2. **The CG rule shapes the whole configuration.** Moving the CG from 17.5 % to a more realistic 35 % of body length costs 19 % drag and shrinks the canards from 114 to 72 mm. The `cg_offset` of 150 mm turns out to be drag-optimal, which suggests it was tuned rather than derived from a mass budget.
3. **Three geometry limits matter, and one is marginal.** The fixed 65 mm pylon length sits just above a cliff (55 mm costs +38 %) and leaves only 1.5 mm between propeller tip and body. The pylon thickness rule is inconsistent with the aerodynamics (+8 to +13 % once fixed). The body diameter requirement moves drag by −5 / +10 %.
4. **Eight of the nineteen numerical assumptions have no effect at this design point** (power, tip Mach, stall, propeller-axis and clearance limits, RPM range, differential-thrust limit). They need no calibration yet.
5. **The optimizer can stop in a worse local optimum** (1 of 8 random starts: +58 % drag). Design runs should always use several starting points.
6. **With this propeller, measured flight power will not calibrate drag:** doubling the drag raises shaft power by only 18 %, because the propeller runs close to windmilling.

1. Purpose and method

Version 1 is being calibrated against a 200 km/h test vehicle before it is developed further. This study asks, for every assumption in the model, three questions: *how much does the answer depend on it*, *is the current value justified*, and *what measurement would pin it down*.

- **Design-level sensitivity.** For each case the full optimization is re-solved, so the design is allowed to adapt. The numbers are the change in *optimal* drag (the objective, baseline 1.308 N); shaft power is also recorded. This is more meaningful than perturbing a frozen design, because it shows what the assumption does to the design you would actually build.
- **Ranges.** Each numerical assumption in `Assumptions` was set to a low and a high value that represent a plausible uncertainty or design alternative (typically ±15–30 %), one at a time. Hard-coded modelling assumptions (airfoils, transition, missing drag terms, CG, atmosphere, objective, resolution) were varied by patching the code inside the study only; the repository code is unchanged.
- **Scope.** 101 optimizations (12 minutes). One-at-a-time variation does not capture interactions; the CG × static-margin interaction, which turned out to be strong, was swept separately (Section 3.1).

Verdicts used in the tables: **Justified**, **Plausible**, **Weak**, **Unjustified**, **Design choice**, **Not active**. *Not active* means the constraint has slack at this design point, so its value does not affect the result.

2. Results at a glance

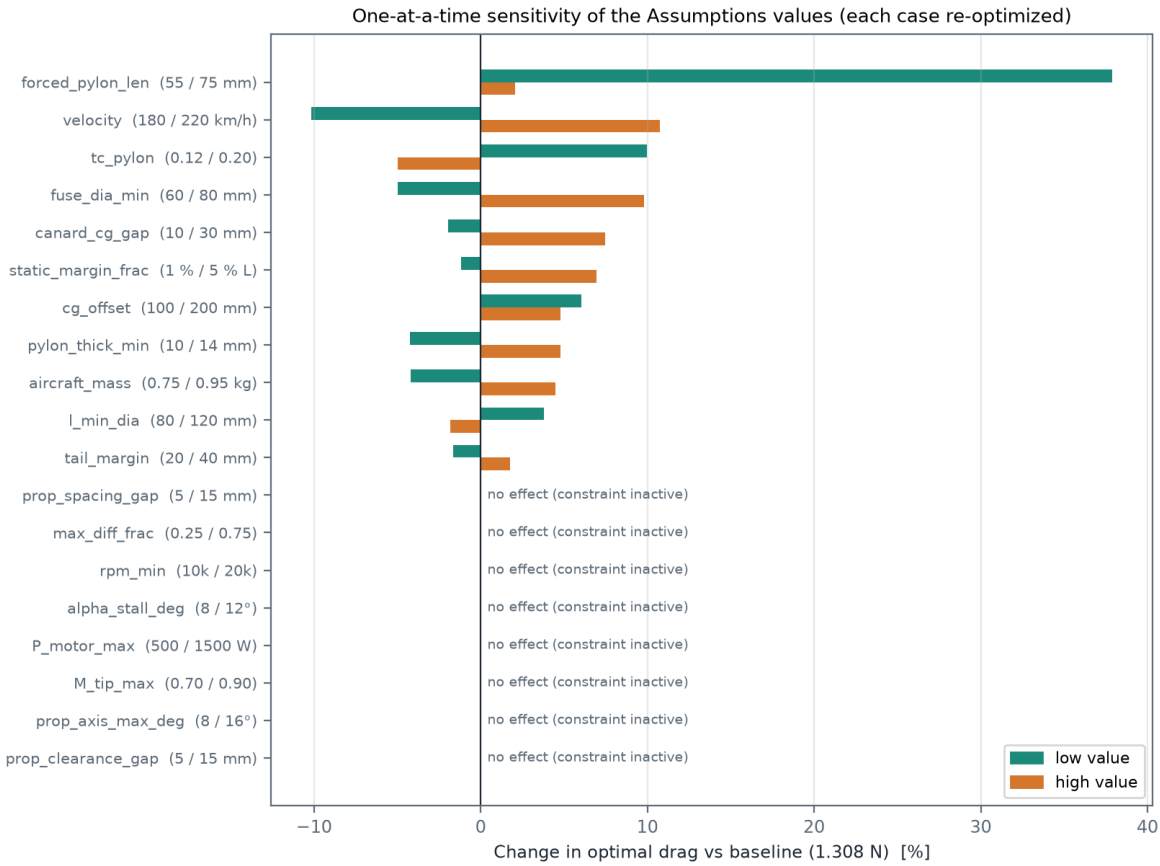


Figure 1. Change in optimal drag when each numerical assumption is set to its low (teal) and high (orange) value. Every case is a full re-optimization.

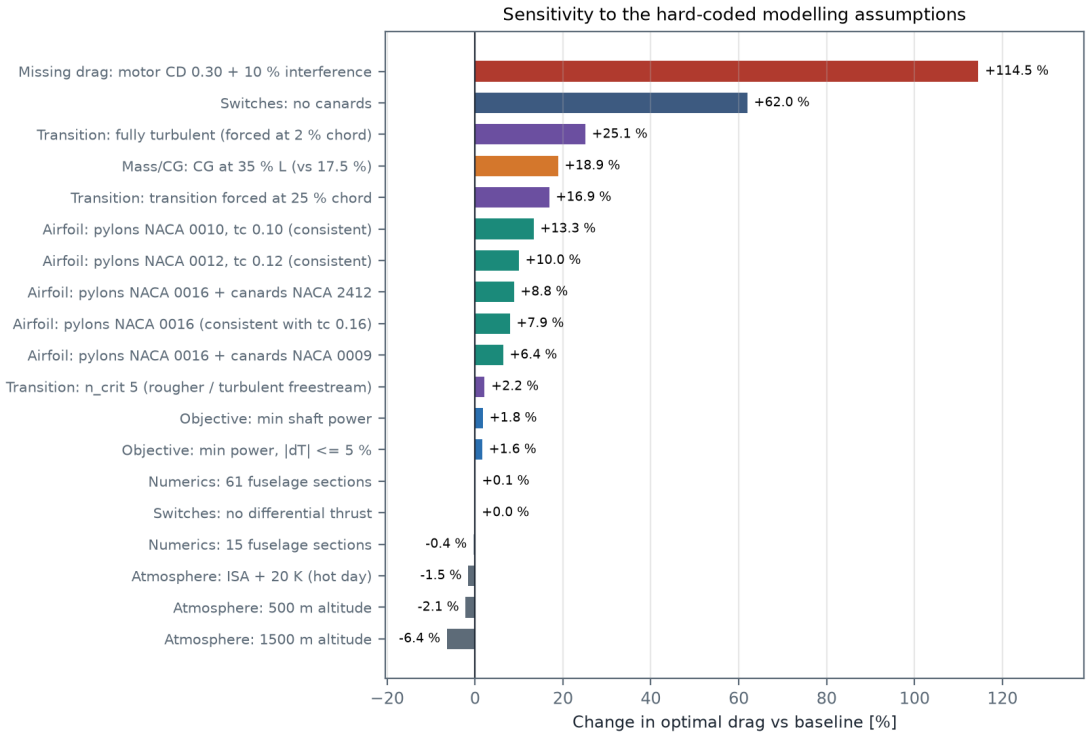


Figure 2. Change in optimal drag for the hard-coded modelling assumptions. "Missing drag" adds four motor nacelles (28 mm diameter, C_D = 0.30) and 10 % interference drag on the profile drag.

3. Assumption by assumption

Each row gives the current value, the drag change over the tested range, a verdict and how to justify or calibrate it. The ranges tested are given in brackets.

Assumption	Value	Effect on drag	Verdict	Reasoning / how to calibrate
Mass and CG				
CG position (free variable, lower bound 50 mm) and `cg_offset`	CG at 17.5 % L; pylons 150 mm aft	CG 25 / 35 / 40 % L: +4.1 % / +18.9 % / +37.7 %; cg_offset (100 / 200 mm): +6.0 % / +4.8 %	**Unjustified**	No mass model: the optimizer puts the CG where it is cheapest. cg_offset is at its drag optimum, i.e. tuned. **Build a mass budget; weigh and balance the prototype.**
`aircraft_mass`	0.85 kg	-4.2 % / +4.5 % (0.75 / 0.95 kg)	**Plausible**	Weak effect. Fixed regardless of geometry. **Weigh the prototype**; later make mass depend on the design.
Stability				
`static_margin_frac`	2 % L (5.7 mm)	-1.2 % / +6.9 % (1 / 5 %); at CG 35 % L: 2→5 % costs +23 points	**Weak**	No reference value for this configuration. Small margins are cheap only because the CG sits at the nose. **Fly a CG sweep to locate the real neutral point** (calibrates $C_{m\alpha}$) and choose the margin from handling.
Geometry limits				
`forced_pylon_len`	65 mm	+37.9 % / +2.1 % (55 / 75 mm)	**Weak**	Sits just above a cliff: shorter pylons need a fatter body to keep the props apart. Leaves only 1.5 mm between propeller tip and body (65 − 63.5 mm). **Check the real clearance; consider making it a variable.**
`pylon_thick_min` / `tc_pylon`	12 mm / 0.16	-4.2 % / +4.8 % (10 / 14 mm); tc: +10.0 % / -5.0 % (0.12 / 0.20)	**Unjustified**	Inconsistent: the aerodynamics use 12 % thick sections. Consistent options cost +8 to +13 % (Section 4.3). **Derive 12 mm from wiring/spar needs; tie the airfoil to t/c.**
`fuse_dia_min` @ `l_min_dia`	70 mm @ 100 mm	-5.0 % / +9.8 % (60 / 80 mm); station: +3.8 % / -1.8 % (80 / 120 mm)	**Plausible**	Strong effect. Must come from the largest internal component (battery cross-section). **Justify from packaging.**
`canard_cg_gap`	20 mm	-2.0 % / +7.4 % (10 / 30 mm)	**Weak**	Purpose undocumented, yet active with a 7 % effect. **State why it exists or remove it.**
`tail_margin`	30 mm	-1.7 % / +1.8 % (20 / 40 mm)	**Plausible**	Small effect. Keep; confirm from motor/ESC packaging.
`prop_spacing_gap`, `prop_clearance_gap`	10 mm, 10 mm	0 %	**Not active**	Spacing has slack; clearance is overridden by the fixed pylon length (see above).
Flight condition				
`velocity`	200 km/h	-10.2 % / +10.8 % (180 / 220 km/h)	**Design choice**	The calibration point. Drag moves ±10 % for ±10 % speed (less than V^2, because the design re-adapts and induced drag falls). Log true airspeed in flight.
Atmosphere	ISA sea level	hot day −1.5 %, 500 m −2.1 %, 1 500 m −6.4 %	**Justified**	Fine for design. **Record test-day temperature and pressure** and use them when comparing with flight data.
Propulsion				
`P_motor_max`, `M_tip_max`, `rpm_min`	1 kW, 0.80, 15 000	0 %	**Not active**	Far from active (465 W total, tip Mach 0.61). Will matter at higher speeds.
APC data, extension to $C_T = 0$	5×3.7 table, $J \le 0.881$	not varied	**Weak**	The optimum runs at $J \approx 0.87$, the edge of the data where it is least reliable. **Measure thrust and power on a thrust stand in a flow at J 0.8–0.9**, or calibrate with a better-matched propeller.
`max_diff_frac` / differential thrust	±50 %	0 % (0.25 / 0.75, and switched off)	**Not active**	The drag objective is indifferent to ΔT, so its value is arbitrary (it flips between −50 % and +50 % across cases). **Set ΔT = 0 at the design point.**

Assumption	Value	Effect on drag	Verdict	Reasoning / how to calibrate
Aerodynamics				
Missing drag: motors, nacelles, interference	not modelled	motor C_D 0.15 / 0.30 / 0.50: +53 / +106 / +177 %; interference +10 %: +8 %	**Unjustified**	The largest effect in the study. **Wind-tunnel test a motor pod and the complete model**, or measure drag in flight from thrust (not power).
Boundary-layer transition	NeuralFoil, free, n_{crit} = 9	n_{crit} 5: +2.2 %; forced at 25 %: +16.9 %; fully turbulent: +25.1 %	**Weak**	Assumes smooth surfaces in calm air; propwash and surface finish will trip the flow earlier. **Oil-flow or tuft test; then fix x_{tr} in the model.**
Airfoils	NACA 0012 everywhere	consistent pylons +8 to +13 %; canard 0009 / 2412: −1.5 / +0.9 % relative to 0016 pylons	**Plausible**	Reasonable placeholder. A cambered canard did not help at this lift coefficient. **Make t/c a design variable.**
Fuselage resolution	31 sections	15: −0.4 %, 61: +0.1 %	**Justified**	Converged.
Formulation				
Objective	minimize drag	min power: +1.8 % drag, −2 W	**Plausible**	At this point both give the same design. Acceptable for calibration.
Canards	on	off: +62 %	**Justified**	Strongly beneficial in this model (while the CG is near the nose).
Local optimum	single start	1 of 8 random starts: +58 %	**Weak**	**Always run several starts** and keep the best.

4. The assumptions that matter most

4.1 Mass, CG and static margin

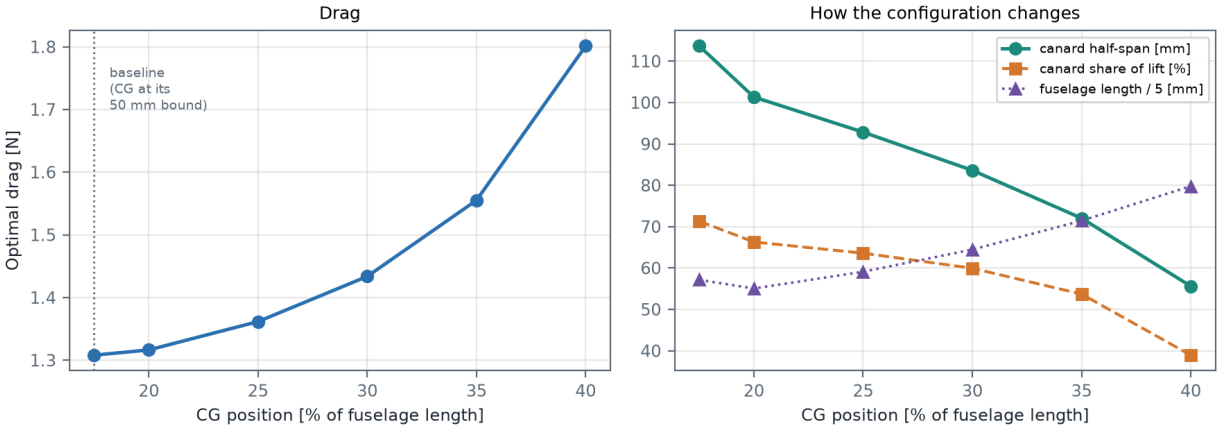


Figure 3. Re-optimized design with the CG fixed at a fraction of the body length (the pylon station is then free). At 45 % L no feasible design exists.

With the CG free, the optimizer moves it to its 50 mm lower bound, and the canard, the only surface near the CG, becomes the main wing (71 % of the lift). Moving the CG aft makes the body longer, the pylons carry more lift and the canard shrinks, but drag rises steadily: +4 % at 25 % L, +19 % at 35 % L, +38 % at 40 % L. The real CG depends on where the battery, four motors and electronics sit, so the mass model is not a refinement: it decides the configuration.

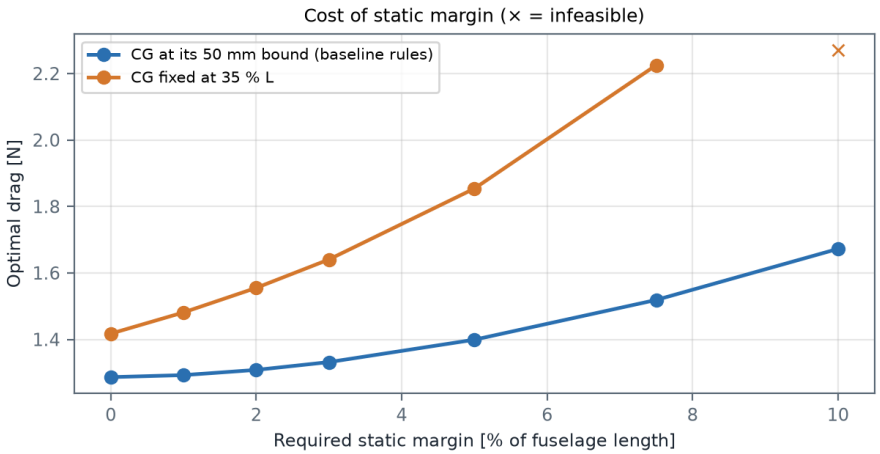


Figure 4. Optimal drag against the required static margin, with the baseline CG rule and with the CG fixed at 35 % L.

Static margin is cheap only because the CG is at the nose. With a realistic CG (35 % L), going from 2 % to 5 % margin costs 0.30 N (+19 %), and 10 % is infeasible. The margin should therefore be chosen together with the mass model, and the neutral-point prediction (AeroBuildup's $C_{m\alpha}$, which includes the destabilizing Munk moment of the body) checked in flight.

4.2 Missing drag and transition

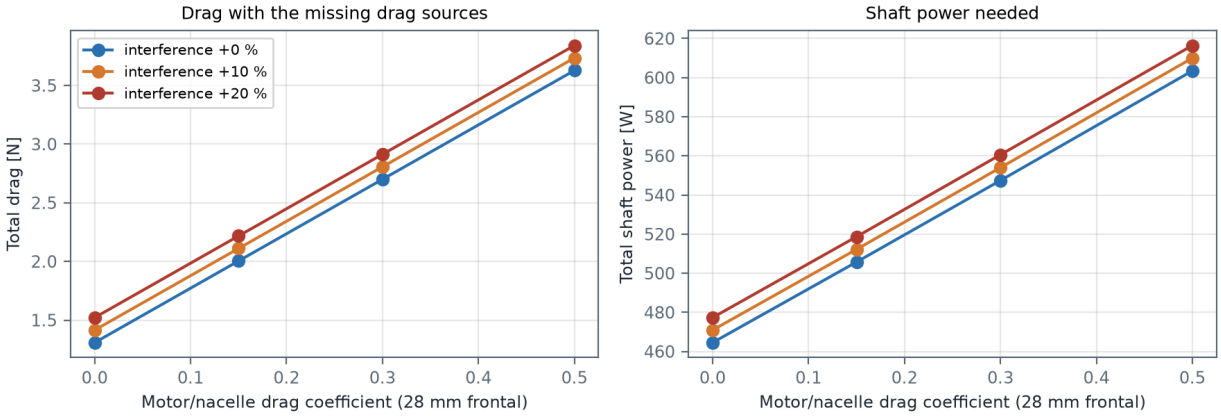


Figure 5. Optimal drag and shaft power when motor-nacelle drag and interference drag are added.

AeroBuildup's drag (1.31 N) contains the body, the pylons, the canards and induced drag only. Four 28 mm motors at the pylon tips, with a drag coefficient between 0.15 (well faired) and 0.5 (bare), add between 0.7 and 2.3 N, i.e. +53 % to +177 %. Forcing turbulent flow adds another 25 %. A realistic total is therefore around 2.5–4 N. These additions barely change the geometry (the drag is added to a fixed set of parts), but they change power, propeller operating point and every downstream sizing decision.

Figure 5 also shows a calibration trap: doubling the drag increases shaft power by only 18 %. The 5×3.7 propeller runs near its zero-thrust point, where most of the shaft power is lost in the propeller itself, so a flight power measurement says little about drag. Drag must be calibrated from a thrust or force measurement, or with a propeller that runs at higher efficiency.

4.3 Pylon geometry

Three assumptions interact at the pylon. (i) The fixed 65 mm length sits just above a cliff: at 55 mm the upper and lower discs can only be kept 137 mm apart by making the body much fatter (+38 % drag). (ii) The same 65 mm leaves 1.5 mm between propeller tip and body, well below the `prop_clearance_gap` of 10 mm the model uses when the length is free; enforcing that clearance (pylon length ≥ 73.5 mm) costs +1.5 % drag. (iii) The thickness rule assumes 16 % sections while the aerodynamics use 12 %. Making them consistent costs +8 % (16 % airfoil) to +13 % (10 % airfoil, longer chord); thicker is better here, because the 12 mm minimum fixes the thickness and a thicker section needs less chord and wetted area.

4.4 What does not matter (yet)

Motor power, tip Mach, stall and propeller-axis limits, the RPM range, the propeller spacing gap and the differential-thrust limit all have slack at this design point, so their values do not affect the result. They will become active as speed rises, so keep them in `Assumptions`, but they need no calibration now. The body resolution is converged, and the atmosphere changes drag by a few percent only.

5. Calibration plan

Ranked by how much each measurement reduces the uncertainty in the model's answer:

#	Measure	Calibrates	Uncertainty it removes
1	Drag of a motor pod and of the complete aircraft (wind tunnel, or flight with an in-line thrust measurement)	missing drag terms, interference, AeroBuildup body and surface drag	−50 to +180 %
2	Mass and CG of the prototype; a component mass budget	CG rule, cg_offset, mass	0 to +38 %
3	Surface flow visualisation (oil flow / tufts) at 200 km/h, in propwash	transition location	0 to +25 %
4	Propeller thrust and power at J 0.8–0.9 (thrust stand in a flow), or a better-matched propeller	APC data near zero thrust	propeller operating point; makes power a usable drag measurement
5	Neutral point from a flight CG sweep	$C_{m\alpha}$, static margin choice	0 to +40 % (via margin)
6	Packaging requirements: battery cross-section, wiring and spar thickness, prop-to-body clearance	fuse_dia_min, l_min_dia, pylon_thick_min, forced_pylon_len	−5 to +38 %
7	Test-day temperature and pressure	atmosphere	a few %

Model changes that would make these assumptions tunable from `Assumptions` (for a later stage): an airfoil family tied to t/c, a transition setting, a parasitic-drag term for motors and junctions, a mass model that returns the CG, a $\Delta T = 0$ option at the design point, and a multi-start wrapper around `solve()`.